

HARSH RAJ

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EDUCATION

Northeastern University , Boston, MA	Dec 2026 (expected)
Master of Science in Computer Science	GPA: 3.8/4.0
Delhi Technological University , Delhi, India	August 2023
Bachelor of Technology in Engineering Physics (Minor in Computer Science)	GPA: 3.4/4.0

TECHNICAL SKILLS

Programming Languages:	Python, SQL, JavaScript, C++, Java
Cloud and DevOps:	AWS, GCP, Docker, Kubernetes, Git, CI/CD
Databases:	PostgreSQL, MySQL, MongoDB, Redis, Vector Databases
Tools and Libraries:	PyTorch, Jax, Hugging Face, Ray, LangChain, Pandas, NumPy, CUDA, Triton

PROFESSIONAL EXPERIENCE

Machine Learning Research Intern, Scale AI , Manhattan, NYC, NY	May 2026 – Sept 2026
- Research on AI Agent Failure Taxonomy. Impact: Released two works on agentic failure modes, covering taxonomy and RCA.	
Research Engineer Intern, Bespoke Labs AI , Mountain View, CA	Jan 2026 – Apr 2026
- Curated synthetic RL training environments for leading frontier labs. Impact: Work contributed to some of the best proprietary CLI agents in the market.	
Research Assistant, Forschungszentrum Jülich	July 2025 – Present
- Contribute to EU Open Foundational Model projects (MINERVA, OpenEuroLLM, ELLIOT) on JUPITER, EU's largest supercomputer. - Lead ablation studies on dataset mixtures and scaling laws on petabyte-scale LLM training data. - Build high-quality permissive datasets for LLMs, with data processing pipelines for mixing, formatting, upsampling, interleaving, and quality validation. Impact: Released MixtureVitae-300BT achieving 10% performance improvement over existing permissive datasets, also matching non-permissive alternatives.	
Applied Scientist, Vijil AI	Aug 2023 – Nov 2024
- Contributed to vijil-evaluation platform for enterprise LLM workflow red-teaming. Integrated 15+ benchmark datasets for vulnerability assessment - Developed vijil-fuzzer framework for adversarial text mutation, achieving 85% success rate against GPT-4o at launch Impact: The vijil-evaluate service is used by enterprise clients including DigitalOcean and Groq	

SELECTED PUBLICATIONS

Model or Harness? An Interaction-Centric Taxonomy for Localizing Agent Failures	1st Author	<i>preprint</i>
Terminal-Bench: Benchmarking Agents on Hard, Realistic Tasks in Command Line Interfaces	Core Team	ICLR 2026
MixtureVitae: Web-Scale Pretraining Dataset With High Quality Instruction and Reasoning	1st Author	ICML 2026
Measuring Reliability of Large Language Models through Semantic Consistency	1st Author	NeurIPS 2022
<i>Best Paper Award</i>		

PROJECTS

terminal-bench (1500+ GitHub ★)	Industry standard Benchmark for CLI Agents. (Core Team)
harbor (1500+ GitHub ★)	Language Agent Sandbox Evaluation framework. (Core Team)
inspect-ai (1300+ GitHub ★)	LLM Evaluation Suite. (Contributor)
garak (5000+ GitHub ★)	LLM Red-Teaming Suite. (Contributor)

AWARDS AND ACHIEVEMENTS

SWISS AI Grant Recipient – Awarded 200k GPU hours and CHF 61,000 in funding	2025
3rd Place – Vision and Language Navigation R2R Benchmark Competition	2024
ML Safety Challenge – Best Paper Award with a \$5000 prize	2022
Rank 7 – Codeforces Raif ML Round 1 Competition	2021